Fixed Point Classification

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Resuming work on Linear Systems[1] regarding the Logistic Map[2] using a Cobweb Plot[3]:

A **fixed point** is a point p such that $f(p) = p$. Usually a sink/source, but due to periodic points, can be a rotation set of points.

> The **epsilon-neighborhood** $N_\epsilon(p)$ of a point p is the internal $\{x \in \mathbb{R} : |x - p| < \epsilon\}$.
>
> If $\exists \epsilon > 0 : \forall x \in N_\epsilon(p)$, $\lim_{k \to \infty} f^k(x) = p$ then we call p a **sink** or **Attracting fixed point**[4]

In plain English, if there is some neighborhood around point p such that all points in that neighborhood converge to p under iteration of f, then p is a sink.

> If $\exists \epsilon > 0$: all $x \in N_\epsilon(p)$ except p eventually map outside of $N_\epsilon(p)$, then we call p a **source** or a **Repelling fixed point**[5].

In plain English, if there is some neighborhood around point p such that all points in that neighborhood (except for p itself) eventually leave that neighborhood under iteration of f, then p is a source.

$_Forcasesinthisclass,\epsilon$ represents a small positive real number parameter.

0.1 Cubic Example

Looking at the function:

$$f(x) = \frac{3x - x^3}{2}$$

Desmos

<iframe src="https://www.desmos.com/calculator/mqr1jsj7l9" width=600 height="400" ></iframe>

The **fixed point sink**[6]**s are**: $x = \pm 1$

The **fixed point Source**[7]**s are**: $x = 0$

[1] https://ewan.my/notes/chaos-theory/linear-systems
[2] https://ewan.my/notes/selfstudy-mathematics/logistic-map
[3] https://ewan.my/notes/selfstudy-mathematics/cobweb-plot
[4] https://ewan.my/notes/chaos-theory/attracting-fixed-point
[5] https://ewan.my/notes/chaos-theory/repelling-fixed-point
[6] https://ewan.my/notes/chaos-theory/attracting-fixed-point
[7] https://ewan.my/notes/chaos-theory/repelling-fixed-point

Key observations:

- Sinks at $x = \pm 1$ are **super-attracting**: $f'(\pm 1) = 0$ (critical points = fixed points)
- Source at $x = 0$ has $f'(0) = \frac{3}{2} > 1$ (linearly unstable)
- **Odd symmetry**: $f(-x) = -f(x)$ creates symmetric basins of attraction
- Topologically similar to Logistic Map[8] at $r = 3$ (pre-chaotic regime)

Theorem

Let f be a smooth map (derivatives continousous), and assume p is fixed point of f.
If $|f'(p)| < 1 \Rightarrow p$ is a sink[a].
If $|f'(p)| > 1 \Rightarrow p$ is a Source[b].
If $|f'(p)| = 1$, we need more information.

[a] https://ewan.my/notes/chaos-theory/attracting-fixed-point
[b] https://ewan.my/notes/chaos-theory/repelling-fixed-point

Origin is a "neutral" fixed point, all other $x \in \mathbb{R}$ are period 2. In essence, some starting position will take a minute to read the period two orbit, whereas others like p_1 will already be on the period two orbit.

0.2 Periodic Sinks and Source

> Let f be a map and assume p is a period k point. The period k orbit of p is a periodic sink (source) if p is a sink(source) for the map f^k

In plain English, if p is a period k point, then we can look at the kth iterate of f and classify p as a sink or source based on that.

[8] https://ewan.my/notes/selfstudy-mathematics/logistic-map